

Leveraging LLMs in Incident Reporting

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Comparing Processing Reported Incident Text with Large Language Models and Untrained Human Screeners



Abstract

We explore the efficacy of large language models (LLMs) in classifying coal mining incidents, using **narrative descriptions** of the event. **Accurate classification** is critical in industries that manage risk as it supports the development of improved mitigation mechanisms. Our study compares the performance of LLMs against **human judgment** via guidelines provided or just label data.

This paper seeks to answer the question: **"How well do LLMs perform as compared to human judgment in a risk classification task?"** By examining both LLMs and human classifiers under the same conditions, our research aims to assess the potential of LLMs to **enhance decision-making** processes in risk-sensitive industries, especially as automation increases both in depth and breadth.

LLM	Human Assessment	
	Near Miss	Minor
Near Miss	534	103
Minor	316	142

Figure 1
Assessment comparison matrix

Deterministic “Temperature”

Most LLMs have a **non-deterministic nature** by design in order to provide different responses for the same prompt. How different the responses are can be parameterized as the **“temperature”** that typically ranges from 0 to 1 (or 2). For example, a calculator would have a temperature of 0.0, but a poet would have a temperature of 1.0. As part of our experiment, we left the default temperature adjustment happen internally in OpenAI’s model [2], but then compared it to the responses with the temperature set to 0.01. This gave more **consistent responses** across querying the same prompt, but did not improve the overall overlap with the human assessments.

Methodology

We began by processing 200,000 incident records from the Mine Safety and Health Administration (MSHA)[1]. These were filtered to exclude incidents that were obviously noteworthy (where days of work were missed). This selection was critical as these remaining records generally reflect either **minor events** or **near misses**. Approximately 60,000 records remained that could either be naturally less dangerous, or purely good fortune. From this subset, 2,000 records were manually classified by up to five **human classifiers**, each equipped to distinguish between incident types based on detailed descriptions. Additionally, we employed queried **LLMs**, specifically OpenAI and Gemma. These LLMs were refined using **prompt engineering** techniques to enhance their response quality and understanding of risk classification criteria. This approach allows us to compare the **effectiveness** and **similarity** of human judgment versus automated models in classifying risk-related incidents.

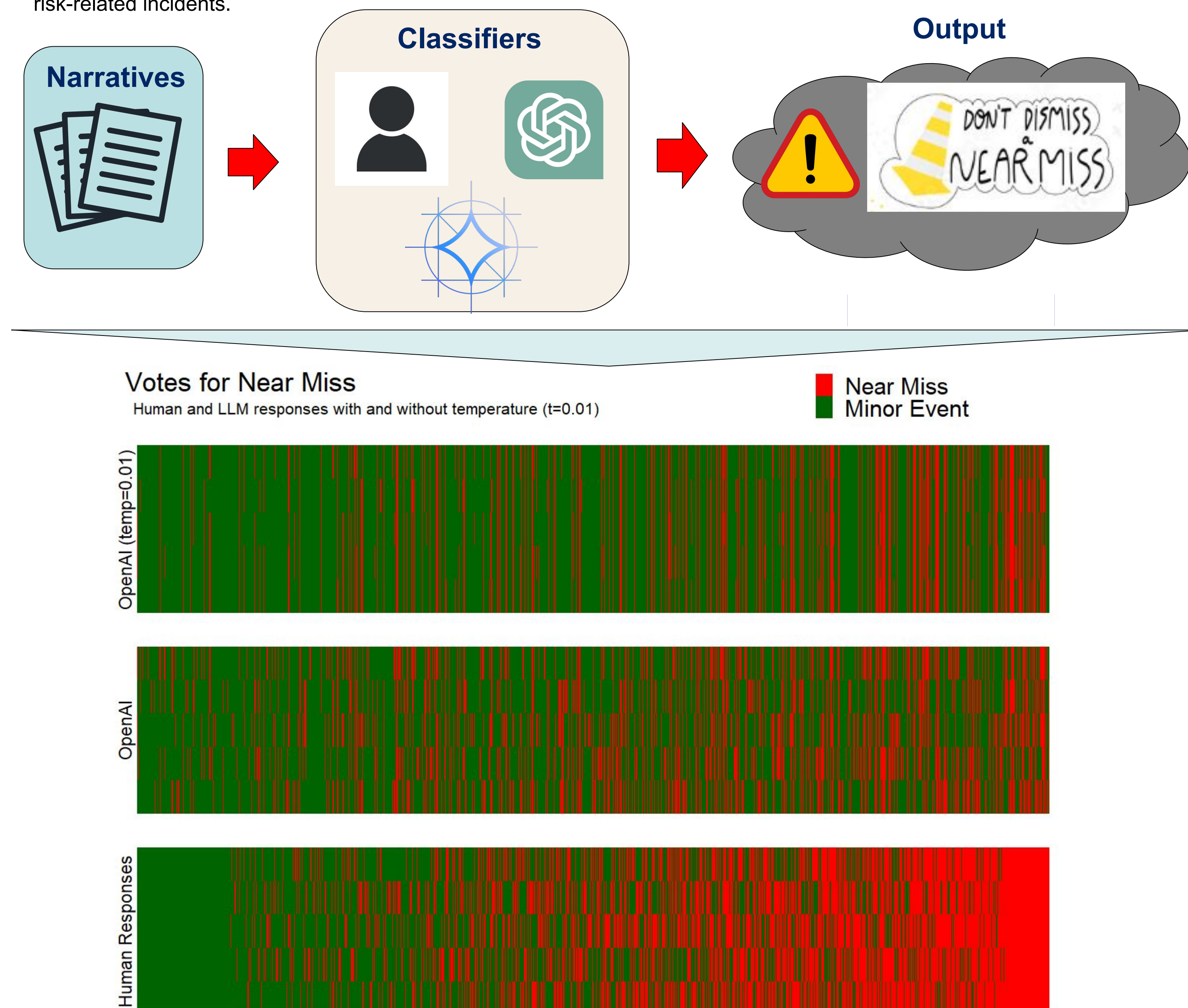


Figure 2
Comparison of responses

Findings & Next steps

- LLMs with an identical instruction prompt cannot **replicate** human-like judgment.
- The temperature parameter helps **identify** and **prioritize** more critical events.
- Better results might be achieved by employing fine-tuning techniques on different models.

Conclusion

While the comparison between human **judgement** and the responses of LLMs is still quite stark overall, the **speed** of LLMs is unmatched. The human responses took weeks to compile, but an LLM can be queried with thousands of prompts in a matter of minutes. Many **parameters**, including temperature, and many other factors about the prompt can also be adjusted to condition the response of LLMs, but humans are much **less deterministic** and disciplined. While much research and experimentation is yet to be done, these results clearly indicate that LLMs in their current state are far from a drop-in replacement for human **assessment of risk**.

Ethical Considerations/Should you worry?

What if PII were to accidentally be publicized in a reporting system?

Replacing someone's job versus giving someone a tool to use in the their current role.

What if thousands of jobs were lost due to actions resulting from reporting?

Potential for bias in the LLMs used to classify risk. Given that these models are trained on existing data, there's a risk they could perpetuate or amplify existing biases in incident classification, potentially affecting safety measures. To address this concern, we should craft better prompts or fine tune our models with diverse datasets. and include a wide range of scenarios.

References

- [1] Mine Safety and Health Administration <https://www.msha.gov/>
- [2] OpeAi <https://platform.openai.com/docs/guides/text-generation/how-should-i-set-the-temperature-parameter>